

National University of Computer and Emerging Sciences



Lab Exercise 11 **AL2002-Artificial Intelligence Lab**

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Exercise

1. Download the dataset given below from kaggle:
<https://www.kaggle.com/datasets/miadul/hospital-readmission-risk-dataset>
2. Perform preprocessing (as covered in Lab 8 and 9) such as:
 - handle missing/duplicate values (remove or impute)
 - Split features and labels (X and y)
 - Split the dataset into 20% testing and 80% training sets.
 - Encode categorical variables
 - Normalize or standardize numerical features
3. Create a simple neural network model given below using tensorflow

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	1,472
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 3)	99

Use `adam` Optimizer and `sparse_categorical_crossentropy` if you performed label encoding on output labels and `categorical_crossentropy` as loss function only if you one-hot encoded the output labels.

4. Train the model using early stopping with the following parameters and analyze performance
 - `monitor='val_loss'`,
 - `patience=5`,
 - `epochs=100`,
 - `batch_size=32` and
 - `validation_split=0.2`

5. Plot learning curves (training vs validation loss). Identify whether overfitting occurs
6. Record test accuracy
7. Now improve the model by adding dropout layers as given in the architecture below:

Layer (type)	Output Shape	Param #
dense_6 (Dense)	(None, 64)	1,472
dropout_2 (Dropout)	(None, 64)	0
dense_7 (Dense)	(None, 32)	2,080
dropout_3 (Dropout)	(None, 32)	0
dense_8 (Dense)	(None, 3)	99

and train this model again with early stopping.

8. Now apply K-Fold Cross Validation to both models (K=5) and answer the following questions:
 1. Did early stopping improve generalization?
 2. How did dropout affect overfitting?
 3. Which model performed best and why?
 4. Compare:
 - Training accuracy
 - Validation accuracy
 - Test accuracy
9. Train any one classification model using Scikit-learn (e.g., Logistic Regression, Decision Tree, or SVM) on the same dataset. Then apply 5-Fold Cross Validation and report the results.
10. Compare the performance of the Scikit-learn model with the TensorFlow models. Which model performed better and why?